

MESSINA, Italy

WMO: 164200

Lat: 38.2005N

Lon: 15.5530E

Elev: 167

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	42.8	45.1	25.9	20.0	49.3	30.4	24.7	51.4	21.5	58.4	19.6	58.3	4.8	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	91.7	75.7	89.3	75.6	87.4	75.5	81.7	86.0	80.2	84.9	78.9	83.9	7.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	159.7	84.4	78.9	150.8	83.4	77.3	143.1	82.4	45.5	85.7	43.8	84.9	42.5	83.9	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.4	14.3	DB	38.8	98.0	2.9	5.2	36.6	101.7	34.9	104.7	33.3	107.6	31.1	111.4
			WB	34.8	83.4	2.4	3.7	33.0	86.0	31.6	88.2	30.3	90.3	28.6	93.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	54.5	54.2	57.3	61.4	68.2	75.9	81.4	82.1	76.0	69.7	62.8	57.0
	DBStd	10.67	4.03	4.34	4.18	3.92	4.19	4.79	3.43	3.24	4.16	3.92	4.32	4.37
	HDD50	32	11	13	3	0	0	0	0	0	0	0	0	5
	HDD65	1348	327	302	239	118	14	1	0	0	0	8	89	250
	CDD50	6160	149	130	231	343	563	777	975	995	779	612	384	222
	CDD65	2001	0	0	2	11	112	327	510	530	329	155	23	2
	CDH74	15742	0	0	6	14	297	2182	5165	5628	2113	323	13	1
	CDH80	4731	0	0	2	1	33	542	1747	1982	403	20	1	0
Wind	WSAvg	6.3	6.4	6.6	6.8	7.3	7.0	6.5	6.5	5.8	5.4	5.5	6.0	6.0
Precipitation	PrecAvg	34.0	4.6	3.7	3.5	2.5	1.3	0.7	0.7	1.1	2.9	4.2	4.4	4.9
	PrecMax	60.1	12.1	9.7	9.2	6.6	8.7	3.1	3.3	5.4	7.1	14.6	10.0	12.8
	PrecMin	18.3	0.7	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.2	1.0	0.6
	PrecStd	6.9	2.5	1.9	2.2	1.6	1.3	0.8	0.8	1.3	1.8	2.8	2.0	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	67.7	71.8	75.6	83.8	94.5	96.2	95.2	89.6	82.3	75.6	70.6
		MCWB	59.3	57.9	59.5	61.1	67.8	76.1	73.3	76.4	76.0	71.4	68.5	63.7
	2%	DB	63.0	63.6	67.8	72.1	80.1	88.0	91.5	91.7	86.2	79.1	72.6	66.3
		MCWB	58.1	57.2	58.4	59.6	67.2	73.6	74.6	76.8	74.6	70.1	65.6	60.9
	5%	DB	61.3	61.6	65.2	69.6	77.2	85.4	89.1	89.4	84.1	77.2	70.5	64.5
		MCWB	56.6	56.0	57.2	59.3	65.8	73.1	75.3	76.2	73.3	69.3	64.4	59.7
	10%	DB	60.0	60.0	63.2	67.6	74.9	82.8	87.2	87.5	82.2	75.5	68.9	62.9
		MCWB	55.3	54.9	56.5	59.2	64.8	72.1	75.5	76.1	73.2	68.6	63.4	58.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	60.9	63.0	66.3	74.5	81.9	83.8	84.1	80.7	75.2	70.6	64.9
		MCDB	63.5	63.8	67.6	70.4	79.1	86.1	87.3	88.2	84.6	77.6	74.1	67.8
	2%	WB	59.2	59.0	60.9	63.8	71.3	78.9	81.4	82.1	78.6	73.1	67.9	62.8
		MCDB	61.7	62.0	64.6	67.9	75.9	84.4	85.7	86.3	82.6	76.7	70.5	65.4
	5%	WB	57.7	57.3	59.4	62.3	68.8	76.8	79.9	80.5	76.9	71.4	66.1	60.8
		MCDB	60.6	60.2	63.0	66.3	73.7	82.3	84.7	85.3	80.9	75.1	69.4	63.6
	10%	WB	56.2	55.9	58.0	61.0	66.9	74.7	78.6	79.1	75.1	69.9	64.3	58.8
		MCDB	59.4	59.2	61.8	65.1	72.2	80.3	83.7	84.2	79.8	74.1	67.9	62.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	9.2	9.9	10.5	10.8	11.1	10.7	10.1	9.3	8.6	7.7
		MCDBR	8.4	9.3	11.2	12.7	13.0	13.3	13.2	12.8	11.5	10.7	9.7	9.0
		MCWBR	6.2	6.3	7.2	8.1	8.4	9.3	9.5	9.7	7.6	6.3	6.8	6.6
	5% WB	MCDBR	7.8	8.6	10.6	11.2	11.9	12.4	12.0	11.6	10.4	9.8	9.0	8.4
		MCWBR	6.7	6.8	7.7	8.2	8.9	9.5	9.5	10.1	8.5	7.3	7.7	7.6
Clear-Sky Solar Irradiance	taub	0.350	0.371	0.412	0.445	0.445	0.428	0.426	0.432	0.436	0.414	0.380	0.355	
	taud	2.454	2.373	2.252	2.176	2.228	2.316	2.340	2.322	2.311	2.333	2.396	2.443	
	Ebn at Noon	261	269	268	265	267	271	270	266	258	250	246	249	
	Edh at Noon	27	32	40	46	44	41	39	39	38	33	28	25	
All-Sky Solar Radiation	RadAvg	676	927	1343	1727	2078	2313	2367	2092	1546	1117	757	601	
	RadStd	55	78	102	115	110	119	72	104	74	85	58	47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.63	N/A	N/A	N/A	N/A

Nomenclature: See separate page